

Lena Eys

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Education

University of Toronto

Toronto, ON

Bachelor of Applied Science (B.A.Sc) Mechanical Engineering + **PEY Co-op** (Sep. 2024 - May 2029)

Technical and Professional Skills

Mechanical: Solidworks (CSWA certified), Trimble Sketchup, 3D printing, composites, soldering, machining

Software: Arduino, Python, MATLAB, OptiTrack, Visual 3D, Minitab, Microsoft Excel, Unity

Professional: Quantitative research, client consultation, DELF B-2 French Certification

Relevant Experience

Laboratory Technician

Wilfrid Laurier University, Neuromechanics Lab

(May 2026 - Aug 2026)

- Independently designed and built a prototype of a flip-up tripping device for use in trip perturbation experiments that could be triggered by a combination of both computer and forceplate input.
- Prepared a Bill of Materials; performed torque calculations for the weight of the tripping barrier, arduino coding, circuit design to breadboard integration, and troubleshooting using a multimeter and continuous design iterations.
- Collaborated with manufacturing companies to ensure parts were delivered within required timelines/tolerances.
- All design documents and prototypes were completed in 75% of the allotted timeframe.

Research Assistant

Wilfrid Laurier University, Neuromechanics Lab

(June 2025 - Aug 2025)

- Analyzed data from in-ground force plates, a load cell and oscilloscope setup, and 3D markers connected to an OptiTrack system.
- Experiments included finding shoe-ground/insole-fabric static and dynamic coefficients of friction, as well as tracking the max moment while a shoe underwent torsion and bending. This allowed the research team to develop a comprehensive set of footwear properties to use in future scientific papers.
- All data was filtered and graphed, and the findings were summarized in a final report.

University of Toronto Engineering Students Consulting Team Member

University of Toronto, UTESCA

(Oct 2025 - April 2026)

- Worked on UTESCA's Haptic Vision project developing a robotic solution to help the visually impaired interact with digital platforms.
- Utilized Python and Unity to convert 2D user interfaces into a digital 3D model, ready for client integration with existing robotic prototypes.
- The full product went on to win the 2026 Troost ILead Difference Maker Award which celebrates making a positive impact in service to others, earning a scholarship of \$50,000.

Science Maker Lab Intern

Wilfrid Laurier University, Science Maker Lab

(Jul. 2024 - Aug. 2024)

- Tutored students in Solidworks design and Arduino coding. Within a week of teaching, they had completed CAD training and were able to code movement patterns for a stepper motor.
- Designed the housing of an introductory Arduino coding box aimed at promoting coding for women at a Laurier WiSTEM event.
- Personal projects included using CAD to design a prosthetic wrist, modeling a storage room for renovations, Arduino custom characters, programming a re-centering motor and automatically activated fan.

Extracurriculars

University of Toronto Varsity Cross Country Member

(September 2024 - present)

- Competed in 6k-8k races with a weekly training commitment of 15-20 hours.

Laurel Creek Track Club competitive member

(July 2022 - present)

- Trained and competed in middle distance events (400m-3000m) and cross country (4-8k) in Canadian provincial and national championships, as well as showcases in Ohio and Michigan.
- U18 Steeplechase provincial gold and Legion Nationals bronze medalist (2023).

Honours and Awards

Leslie & Lois Shaw Admission Scholarship (Oct. 2024) - \$3,110

150th Anniversary Scholarship, KCI Convocation (June 2024) - \$350

Edna Haviland Award for Excellence in Chemistry, KCI Convocation (June 2024) - \$175

Dr. Mabel Dunham Award, KCI Convocation (June 2024) - \$1000

Avery Wong

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EDUCATION

University of Toronto

Toronto, ON

Bachelor of Applied Science in Mechanical Engineering + PEY Co-op

Sept 2024 - May 2029 (expected)

Minor in Engineering Business

Relevant Coursework: Engineering Strategies & Practice I, II (Project Management), Statistics (Minitab), Mechanical Engineering Design (Solidworks), Manufacturing (Basic Machining), Numerical Methods (MATLAB), Computer Programming (Python)

SKILLS

Engineering & Programming Tools: SolidWorks, ANSYS, MATLAB, Python, Minitab, HTML/CSS

Productivity Tools: Google Workplace (Sheets, Slides), Microsoft Office (Excel, PowerBI, Project, Powerpoint)

Professional Skills: Project Management, Communication, Collaboration, Troubleshooting, Adaptability

EXPERIENCE

Mechanical and User Experience Lead

Toronto, ON

University of Toronto Biomedical Design Team

Aug 2025 – May 2026

- Led research and development (R&D) efforts to create a smart rehabilitation knee sleeve that tracks muscle activation and range of motion in users that are recovering from ACL tears
- Applied CES EduPack software and market benchmarking to identify and filter candidate materials based on multiple performance criteria, shortlisting materials for further prototype testing and analysis
- Designed and modeled geometric prototypes using SolidWorks and conducted physical tests with sample materials and stress analysis simulations through ANSYS to determine final material composition, resulting in a triple-layered material mesh that balances cost and performance
- Troubleshoot poor surface electromyography (sEMG) signals by increasing knee sleeve compression, optimizing power input, reducing electromagnetic interference, and introducing a grounding electrode, resulting in a 50% improvement in signal quality.

Engineering Lead

Toronto, ON

Mechanical Engineering Design - University of Toronto

Sept 2025 – Dec 2025

- Spearheaded the design and development of a low-cost, long reach 6-DOF robotic arm for cinematography applications
- Facilitated weekly team meetings to assign tasks and set project timelines using Trello and Google Workspace, ensuring on-schedule task delivery across the team
- Designed and modeled a fully constrained digital prototype using SolidWorks, guided by a 1 m/s speed and 5kg payload requirement, validating technical feasibility
- Converted the digital prototype into detailed SolidWorks assembly drawings and Bills of Materials (BOM) across multiple subassemblies, specifying critical dimensions for non-standardized components to facilitate future machining

Team Lead

Toronto, ON

Engineering Strategies & Practice II - University of Toronto

Sept 2024 – May 2025

- Led a team of 6 to conceptually design and develop an electrical outlet extender suited for use in an elementary school classroom under strict time constraints, successfully delivering a prototype integrated desk outlet for elementary classrooms
- Conducted market research and applied ISO Standards and ESA Codes to generate over 75 unique design solutions, ensuring full compliance with client and stakeholder requirements

- Facilitated weekly team meetings to assign tasks through a Gantt Chart and update the project timeline using Microsoft Project, ensuring all milestones were delivered on schedule
- Co-authored an IEEE-formatted technical report and PowerPoint presentation, successfully demonstrating how the proposed design met all the required specifications

ADDITIONAL

Certifications & Training: Certified SolidWorks Associate (CWSA), Basic Machining at George Brown College

Websites: <https://avery-06.github.io/portfolio/>